

Towards Full Body Co-Embodiment of Human and Non-Human Avatars in Virtual Reality

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ABSTRACT

Co-embodiment, which enables users to share virtual avatars with other entities such as users, robots, or autonomous agents, has raised interest in the Virtual Reality community over the past five years. Most co-embodiment studies and implementations only relied on human avatar representation and provided upper-body shared control with the avatar and the users remaining seated. In this work-in-progress paper, we present two new designs of co-embodiment for two users: (1) a full-body shared human avatar and (2) a non-human avatar with four limbs and two heads. We describe the preliminary implementations of our approaches and discuss the potential application and research areas of co-embodiment.

Keywords: Virtual Reality, Dissimilar Avatars, Co-Embodiment

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction Paradigms—Virtual Reality;

1 INTRODUCTION

When Virtual Reality (VR) applications provide a user with self-representation, it is often a partial avatar made of a head and a pair of hands. This choice is usually based on practical considerations of the limited number of tracked points of the user's own body, the motion of which can be mapped to the avatar for a plausible animation. In rarer cases, when a full-body avatar is used, it is often a human or human-like body with a one-to-one mapping of the movements of the user's body parts to those of the avatar. Indeed, this type of embodiment is familiar, convenient, and well-suited for most applications. However, the power and wonder of VR as a fundamentally different medium is precisely in its ability to surpass the conventions of the real world and provide an experience that is genuinely dissimilar to the usual expectations of being and acting. Here, we discuss a way of possessing and controlling a body that is impossible in the real world, referred to as co-embodiment. According to Fribourg et al., co-embodiment is a situation in which a user shares a virtual avatar with another entity, such as another human user or a virtual agent [4]. At this point, co-embodiment is a novel area of VR, with only a few papers addressing the concept. However, it is closely connected to the study of joint action through the element of agency central to embodiment and collaborative interaction. In this work-in-progress paper, we address two facets of co-embodiment. First, we discuss our ongoing research on the full-body co-embodiment of a human avatar. We propose our shared control scheme for the complete avatar animated with a 6-point inverse kinematics (IK) method and discuss challenges posed by locomotion. Second, we propose a novel co-embodied non-humanoid avatar with more body parts one user could comfortably control. Finally, we discuss possibilities for the future of co-embodiment research, current challenges, and potential application areas.

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2 RELATED WORK

It is impossible to discuss co-embodiment without outlining what we know about the much more researched underlying concept, embodiment, which refers to the "subjective experience of using and having a body" [2]. In the more specific context of VR, the Sense of Embodiment (SoE) describes a range of sensations associated with being inside, owning, and controlling a virtual body. It is generally divided into three components: self-location, body ownership, and agency [9]. Self-location refers to feeling inside a virtual body, while the sense of body ownership is defined by attributing the virtual body to oneself. The sense of agency arises from the perceived sense of motor control over the virtual body, the ability to move it at will. It is natural to expect that the sense of agency is the most affected component of SoE in co-embodied scenarios, where the actual motor control of one avatar is shared between users. A meta-analysis showed that in joint actions without co-embodiment, the sense of agency can be experienced over both own and partner's actions [23]. This experience depends on whether predicted and actual outcomes of the action coincide, in line with previous results explaining the sense of agency [3, 18].

The term co-embodiment was first used in the context of physically embodied voice assistants when two such assistants were embodied as the same car in a user enactment study [15]. Fribourg et al. were the first to study the co-embodiment of a virtual avatar by two users, focusing on the perception of agency while sharing control over the avatar's arm [4]. The study found a positive correlation between the degree of control over the avatar and the sense of agency experienced by the user. Supporting the results of Fribourg et al., another study investigating co-embodiment in a seated human avatar with shared control of one arm found that the sense of agency experienced by participants was higher when they were embodied in the solo condition than in the co-embodied condition [6]. However, target reach deviation was improved in the co-embodied condition compared to the solo condition, suggesting co-embodiment potential in applications for motor skill learning. Hapuarachchi et al. studied SoE in a co-embodiment scenario in which each co-embodied user fully controlled one avatar arm, one left and the other right. At the same time, they received haptic feedback when the non-controlled arm was moved through a brace setup connecting their backs [7]. Even in the total absence of motor control of one arm, the haptic feedback positively impacted the sense of body ownership and agency towards the non-controlled arm.

Collaborative training and motor skill transfer are suggested most frequently when applications of co-embodiment are discussed. Co-embodiment of a virtual hand improved subjective usability ratings and reduced time delay during a hand movement imitation task [8]. Kodama et al. proposed a mechanism for motor skills learning, according to which the weights distribution between the teacher's and the learner's co-embodied control changes with the progress of the learner [11], demonstrating a lower performance drop, albeit also lower learning efficiency, after the termination of the co-embodied phase. The same authors reported learning efficiency improved when the learner was co-embodied with the teacher compared to learning alone or simply sharing the teacher's first-person perspective [12]. Co-embodied learning also positively influenced procedural and declarative memory during motor skills learning [22].

Research studies described above-addressed co-embodiment with human avatars, which are not the only possible choice of embodiment in virtual environments. It has been demonstrated that users can learn to control and experience SoE towards avatars with body schemes dissimilar to a human to different extents, from a human avatar with a very long arm [10], an avatar with a third arm [20], an avatar with a tail [19] to completely non-human avatars such as a rhino, a scorpion and a bird [13, 14]. These avatars have not been tested in co-embodied scenarios. When a single user embodies an avatar with a different body schema or additional body parts, the control of the avatar can be achieved by modifying the tracked movements of the user's body parts to adapt them to the dissimilar schema before using them to animate the avatar [20].

3 NOVEL CO-EMBODIMENT DESIGNS

3.1 Full-body human avatar co-embodiment

While previous work on co-embodied human avatars focused on shared control of avatar arms and generally investigated seated scenarios, we also propose a novel shared control mechanism for the lower body. The result is full co-embodiment, allowing users to stand, walk around, and perform arbitrary movements. Below, we describe the design principles of the proposed shared avatar control and then give details on our specific implementation. We consider several principal body parts of the avatar, for which position and rotation need to be calculated based on the input of two users: head, torso, arms, and legs. Besides these body parts, the co-embodiment mechanism must consider the avatar's global position and scale.



Figure 1: Two users and the avatar they co-embodiment, as seen in the mirror by the left user. The colored spheres show the relative positions of the hands and feet of the other user. Up: averaged arms pose; middle: averaged legs pose; bottom: averaged torso pose.

Control weights. Our design allows us to set weights for adjusting the avatar control between two users. The weights assigned to both users add up to 100%. Several weight assignment modalities are available: setting the same weight distribution for all body parts, customizing weights for each body part, and splitting the control of

the avatar between two users along the vertical or horizontal axis. Fig. 1 shows an avatar embodied with the equal, 50-50% weights.

Global position and scale. To prevent disruptions in SoE that could result from the avatar being too large or too small for a user, we scale the co-embodied avatar to the user's height in each user application. This way, the avatar's size is not synchronized over the network. To prevent any possibility of cybersickness, the control of the virtual camera in each client application is not shared with the other user and is determined by the tracked pose of the head-mounted display (HMD). The global position of the avatar (its local copy) also follows the tracked pose to prevent out-of-body situations.

Head. In single-user embodiment, the pose of the avatar's head is usually determined by the pose of the HMD, with the virtual camera position inside the avatar coinciding with the position of the eyes. We keep to the same convention since the pose of the virtual camera is not shared; however, we apply weighted head rotation around the up-axis to prevent unnatural head rotations with respect to the torso.

Torso. Since users can move in the world space independently of each other, the absolute position of the torso in the horizontal plane is not shared and is determined by the tracking data alone. However, since the waist is the central point of the avatar that determines the animation of the feet, its vertical position needs to be weighted to prevent situations in which the feet of the shared avatar would go up in the air if one of the users is standing. The other one is squatting, for example. The rotation of the torso around the up-axis is also weighted to prevent unnatural poses if the rotations of the co-embodied users are too different.

Arms and legs. The poses of arms and legs are calculated relatively to the waist by applying the co-embodiment weights to local relative positions and rotations or hands and feet end-effectors.

Our co-embodiment application is implemented in Unity and uses HTC Vive and Vive trackers to track the users' body parts and animate the avatars. The implementation is based on AvatarGo, a multi-user project that uses 6-point IK to animate avatars [17]. Both users wear the HTC Vive HMD, hold Vive controllers, and have HTC Vive trackers attached to their lower backs and feet to obtain poses of end-effectors for the IK implementation.

In the base implementation of AvatarGo, each user performs a calibration procedure after being connected to the networked scene, in which the scale of the avatar is adjusted. When both users are calibrated, their avatars become visible to each other. The tracking data of the 6 points is synchronized across the network and is used to set the poses of end-effectors in the IK algorithm. In our application adapted to enable co-embodiment, end-effector poses are calculated according to the above rules and not merely replicated from the tracking data. The other player's avatar (its remote copy) is made invisible. This way, the local copy of the self-avatar becomes the co-embodied avatar.

3.2 Co-embodied non-human avatar

For our novel work on non-human co-embodiment, we intended to design an avatar that would benefit from shared control by two users. In particular, we were required to utilize tracked poses of two HMDs and two pairs of controllers to animate separate body parts. Our non-human avatar resembles a partially upright standing slug with two eye stalks and two pairs of tentacle arms protruding from the upper torso. Its default pose is standing upright with eye stalks stretched upwards, and tentacles stretched to the sides.

The character model is animated using a rigged skeleton. Six bones control the torso, and the tentacles are controlled by nine bones each; one of the bones serves only as an anchor and does not move. Each of the tentacles is connected to the torso by a fixed offset. Another chain of bones, similar to a spine, is used to control the lower body. We use four protruding bones connected to the spine chain to stabilize the lower torso. Another chain of five bones can be used to control the tail. The character is animated using IK, with

end-effectors placed at the ends of the tentacles and the eye stalks. The character's default pose, control points, and the rig model can be seen in Fig. 2.

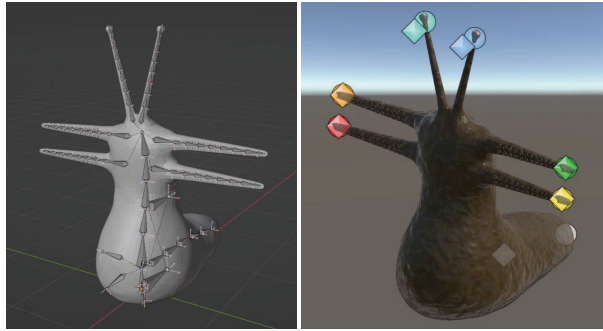


Figure 2: Left: Unshaded model of the slug character with its rig. Right: The slug avatar with all its control points. The circles represent the targets, and the squares represent the pivots.

The shared control of the slug avatar is accomplished by driving the IK control points with the tracking input of two HMDs and four VR controllers. We adapted the FABRIK IK algorithm [1] to make sure that joint and bone rotations are correct for our character design. The two co-embodied users control one eye-stalks and one pair of tentacle arms. The position and rotation of each eye coincide with that of each user's virtual camera and are set to the tracked pose of the HMD. When controlling the eye-stalks together, users can bring them in the pose of looking "at each other". The poses of a user's VR controllers are applied to one pair of tentacles, with one user controlling the upper pair and the other on the lower pair. Two different variants of avatar control are available, "shared controls" and "single controls". In the single controls mode, the user chooses which eye-stalk and tentacles to control and can switch from one pair of tentacle arms to another. In the shared mode, the co-embodied avatar controls described above are applied.

We have implemented a motion scheme for the slug avatar, designed to move its body forward. The global movement of the avatar results from the combined movement of both users in the tracked space. The morphology of the avatar head with the eye-stalks presumes the possibility of the eyes moving around on the stalks locally, with the trunk of the body standing still. Therefore, the movement of the HMDs should not automatically lead to the movement of the slug body. Our solution is a method for avatar forward movement that happens in discrete strides: first, the eye-stalks move forward, following the motion of the users' HMDs, until a specific limit; when this limit is reached, the body of the slug is pulled forward to catch up with the eyes. Each eye exhibits a separate pull on the character's body when it moves away from the torso.

Four zones around the slug's upper torso are defined to implement this global motion behavior as shown in Fig. 3. The blue zone is the idle zone: whenever the eyes of the avatar are within it, the avatar body is still. The green (outermost) zone marks the border at which the forward movement of the slug body starts. As soon as at least one of the eyes crosses the zone's border, the slug's body rotates and moves to the closest location between both eye-stalks in which both eyes (driven by the HMDs tracking input) are not inside the green zone anymore. The red zone triggers a re-alignment of the slug body such that both eye-stalks have the same angle relative to the forward vector. Finally, the yellow zone serves to smooth out the transition between the still and moving states of the body. Suppose at least one of the avatar eyes enters the yellow zone. In that case, the movement of the body is affected depending on in which zone the eye has been previously: if it transitions from the red zone, the motion rules of the red zone apply; if it is in the green zone, the

body keeps moving further until the eye has crossed the border into the red zone to trigger a new re-alignment step.

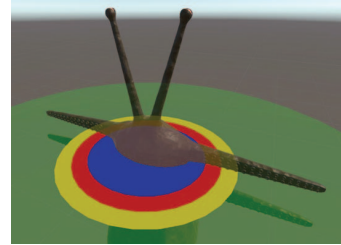


Figure 3: A visual representation of the movement borders and the resulting movement zones (blue: idle-zone, red: stop moving, yellow: move, green: start moving).

4 DISCUSSION

4.1 Limitations and Future Work

While our avatars are the first proof of concept of full-body co-embodiment in VR, we are aware of their limitations. First, the range of possible movements the avatars can achieve is somewhat limited. For the human avatar, our implementation works best when users are similarly oriented; global orientations that are too different, such as back-to-back, can yield unrealistic poses. Similarly, forward motion is adapted only where users walk in the same direction. However, there is currently no way to ensure that co-embodied steps during walking look plausible. In addition, learning how to use the avatar and control it with another user is challenging. Our current implementation presents visual feedback only of the shared avatar pose and of the positions of the partner's hands and feet represented by spheres. The user might feel that the avatar is not responding to their movements, leading to poor agency. Future work should address these challenges to design plausible and compelling locomotion for co-embodied avatars while maximizing SoE.

Regarding the non-human avatar, walking can also be challenging in some situations. For example, if users walk in opposite directions, the slug body would remain still due to the global pull summing up to null. However, the virtual camera may separate from the eyes, leading to out-of-body experiences since we do not freeze its position. We will consider how to address this issue in the following design iterations. One solution might be to extend and deform the avatar model to provide feedback about each user's global position.

Finally, these avatars have not been evaluated yet. We will conduct two user studies to investigate factors influencing the SoE and user interaction with these avatars. These studies will consider each avatar's design space, and a third one will compare human and non-human avatars. In the first study, we will assess the usability of the full-body human avatar during a task requiring upper and lower body coordination (e.g., using a mixed rhythm game combining Beat Saber and Dance Dance Revolution) and determine whether control weights influence user behavior and performance. In the second study, we will explore how users coordinate to interact depending on the body segment assignments (e.g., one user controlling the two left limbs and the other the right ones, or one user controlling the upper ones and the other the lower ones). Eventually, a third study based on the previous two results could compare how users interact in VR with the human or non-human shared avatar.

4.2 New Applications and Research Areas

While current research shows that co-embodiment would benefit training and rehabilitation applications, it still requires further investigation. Co-embodiment could be beneficial as a tool for boosting

mental health, with non-human avatars used as therapeutic interventions for depression and anxiety disorders. Shared control of an avatar could benefit people with disabilities that restrict the range of motion, giving them greater power over controlling an avatar as proposed in [7], but also promoting different bodies and generating empathy. For these possible applications to use the idea of co-embodiment to its full potential, further research on shared agency, the sense of self in co-embodied situations, and the leader-follower dynamics needs to be conducted (e.g., by using production techniques that enable to drive a participant with controllers [16,21]). Moreover, we need to understand how to tailor the individual control over co-embodied avatars to achieve the strongest sense of agency and to what extent co-embodied users need to be aware of each and have a sense of shared goals in different applications. Thus, we suggest that hyperscanning studies (i.e., neuroimaging multiple people at the same time) could provide insights into users' synchronization while interacting with co-embodied avatars as it was shown for eye movements with avatars [5].

Our non-human slug avatar exhibits human-like characteristics, standing upright or semi-upright (depending on whether we consider the tail), displaying symmetry along the vertical axis, possessing two localized viewpoints, and featuring tentacles morphologically similar to human arms. Such design prioritizes simplicity for co-embodiment between two users. However, the concept of co-embodiment with inherently non-human body schemas (e.g., spiders, octopuses, and myriapods) remains an open question. Although anecdotal evidence suggests it is possible [20], there is a lack of research on such topics. Could we extend the concept of co-embodiment to encompass fantastical creatures that do not even have centralized body segments (e.g., having distributed vision or being able to use the whole body as a hand)? Previous research proposed to control additional body parts of dissimilar avatars by using modified tracking data of various user body parts. In contrast, our solution combines the tracking data of two users to drive one avatar. Both approaches may be considered as future research may contemplate even more dissimilar and complex avatars.

5 CONCLUSION

Virtual co-embodiment unlocks new possibilities for immersive experiences and collaboration between humans and virtual agents. While this research topic is new, it is receiving more and more attention in the VR community. We presented preliminary designs of new forms of co-embodiment that aim to provide a broader range of movements than the current ones. Our full-body co-embodiment designs enable interactions between two users who can walk independently in a collocated workspace. We believe that such avatars could improve the understanding of human behavior in VR, including the motor learning process and the sense of agency during joint actions, and could benefit applications such as rehabilitation, training, and entertainment.

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